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İzmir University of Economics

CE 326 (306) Computer Networks

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Lab Exercise #9

In this lab, you will explore error detection mechanisms used in the data-link layer, specifically Parity Bits and Cyclic Redundancy Check (CRC). You will implement these techniques using Java and analyze the generated network traffic using Wireshark.

In the first part of the lab, you will study how parity bits are used to detect errors in transmitted data. You will implement methods to generate data, calculate parity, and append parity bits. In the second part, you will focus on CRC, implementing polynomial division and receiver-side validation.

The goal of this lab is to strengthen your understanding of error detection techniques and develop practical skills in implementing and analyzing these mechanisms through packet inspection.

Academic Honesty and Submission Policy: **All students are expected to adhere to the principles of academic integrity. You must complete and submit your own original work. Any form of plagiarism, copying, or unauthorized collaboration is strictly prohibited and may result in disciplinary action in accordance with university regulations. Submissions must be made individually and during the designated lecture hours. Work submitted outside of the scheduled lecture session will not be accepted.**

Do the following:

- For the Parity Bit part:
  - Download `parityserver.jar` and `ParityBitDemo.java` from Blackboard.
  - Create a new Java project and add the given files.
  - Start the server using: `java -jar parityserver.jar`
  - Implement the following methods in the Main class:
    - `createData(int[] data, int size)`
    - `calculateParity(int[] data, int numOfOnes)`
    - `addParityBit(int[] data, int numOfOnes)`

- For the CRC part:
    - Download `crcserver.jar`, `CRC.java`, and `CRCDemo.java` from Blackboard.
    - Add the classes into your project structure.
    - Start the server using: `java -jar crcserver.jar`
    - Implement the following methods in the CRC class:
        - `divide(int old_data[], int divisor[])`
        - `receive(int data[], int divisor[])`
- Packet Analysis with Wireshark:
    - Open Wireshark and select “Adapter for loopback traffic capture”.
    - Start capturing packets while your client application communicates with the server.
    - Analyze how data is transmitted and observe differences between parity bit and CRC mechanisms.

Please submit your solution as a .ZIP file containing all required Java source files. Make sure your code compiles before you submit.